

CSA0502 – QUERY PROCESSING FOR DATA SCIENCE

Lab Solutions – Set 5

Question 9

Write a Pandas program to create a Pivot table and find the total sale amount region wise, manager wise, and salesman wise.

Python Code:

```
import pandas as pd

# Create realistic sales dataset
data = {
    'Region': ['East', 'Central', 'Central', 'West', 'East', 'Central', 'West', 'East'],
    'Manager': ['Martha', 'Hermann', 'Hermann', 'Timothy', 'Martha', 'Hermann', 'Timothy', 'Martha'],
    'Salesman': ['Alexander', 'Shelli', 'Sigal', 'Stephen', 'Diana', 'Luis', 'John', 'Karen'],
    'Item': ['Television', 'Home Theater', 'Television', 'Cell Phone', 'Home Theater', 'Television', 'Home Theater', 'Cell Phone'],
    'Units': [95, 50, 36, 27, 60, 75, 32, 60],
    'Unit_Price': [1198.0, 500.0, 1198.0, 225.0, 500.0, 1198.0, 500.0, 225.0],
    'Sale_Amount': [113810.0, 25000.0, 43128.0, 6075.0, 30000.0, 89850.0, 16000.0, 13500.0]
}

df = pd.DataFrame(data)

# Create Pivot table to find total sale amount region wise, manager wise, salesman wise
pivot_sales = pd.pivot_table(df, index=['Region', 'Manager', 'Salesman'],
                              values='Sale_Amount', aggfunc='sum')

print("Pivot Table - Total Sale Amount Region-wise, Manager-wise, Salesman-wise:")
print(pivot_sales)
```

Output:

```
Pivot Table - Total Sale Amount Region-wise, Manager-wise, Salesman-wise:
                                     Sale_Amount
Region Manager Salesman
Central Hermann Luis              89850.0
                Shelli             25000.0
                Sigal              43128.0
East    Martha  Alexander         113810.0
                Diana             30000.0
                Karen             13500.0
West    Timothy John             16000.0
                Stephen           6075.0
```

Question 10

Create a DataFrame of 10 rows and 4 columns with random values. Write a Pandas program to highlight negative numbers in red and positive numbers in black.

Python Code:

```
import pandas as pd
import numpy as np

# Create DataFrame of 10 rows and 4 columns with random values
np.random.seed(42)
df = pd.DataFrame(np.random.randn(10, 4) * 5, columns=['Col1', 'Col2', 'Col3', 'Col4'])

# Function to highlight negative numbers red and positive numbers black
def color_negative_red(val):
    color = 'red' if val < 0 else 'black'
    return f'color: {color}'

# Apply styling to DataFrame
styled_df = df.style.applymap(color_negative_red)

print("DataFrame with Random Values:")
print(df)
print("\n(Note: In Jupyter Notebook or VS Code, styled_df renders negative numbers in red
and positive numbers in black)")
```

Output:

```
DataFrame with Random Values:
   Col1    Col2    Col3    Col4
0  2.483571 -0.691322  3.238443  7.615149
1 -1.170767 -1.170685  7.896064  3.837174
2 -2.347372  2.712800 -2.317088 -2.328649
3  1.209811 -9.566401 -8.624589 -2.811438
4 -5.064156  1.571237 -4.540120 -7.061519
5  7.328244 -1.128882  0.337641 -7.123741
6 -2.721914  0.554613 -5.754968  1.878490
7 -3.003193 -1.458469 -3.008533  9.261391
8 -0.067486 -5.288555  4.112725 -6.104218
9  1.044318 -9.798351 -6.640930  0.984306

[Styled DataFrame: Negative numbers highlighted in red, positive numbers in black]
```